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STEP 17 — PROBLEM-SOLVING PORTFOLIO

IT Troubleshooting Cases

Introduction

This document presents five common IT troubleshooting cases and demonstrates a systematic approach to solving technical problems. Each case identifies the problem, possible causes, diagnostic procedure, solution, and lesson learned. The troubleshooting process focuses on identifying the cause before applying a solution rather than making assumptions or changing multiple things at once.

Case 1 — Computer Cannot Connect to Wi-Fi

Problem	The computer cannot connect to the available Wi-Fi network, or it connects but cannot access the internet.
Possible Causes	• Wi-Fi adapter is disabled. • Incorrect or outdated network adapter driver. • Incorrect IP configuration or DNS settings. • Router or access-point problem. • Weak signal or interference. • Temporary network configuration error.
Diagnostic Procedure	1. Check whether Wi-Fi is enabled and confirm that the correct network is selected. 2. Check whether other devices can connect to the same network. 3. Restart the computer and router if appropriate. 4. Check the network adapter status and driver in Device Manager. 5. Use Windows network troubleshooting tools and inspect the IP configuration. 6. Test connectivity using the gateway and an external website to determine whether the problem is local, router-related, or DNS-related.
Solution	Enable the adapter, reconnect to the correct network, restart the router if necessary, update or reinstall the network driver, and correct the IP/DNS configuration when required. If other devices also cannot connect, investigate the router or internet service.
Lesson Learned	Troubleshooting should start by determining whether the problem affects one device or the entire network. This prevents unnecessary changes to a computer when the actual problem is the router or internet connection.

Case 2 — Computer Is Running Very Slowly

Problem	A computer takes a long time to start applications, responds slowly, or becomes unresponsive during normal use.
Possible Causes	• Too many startup applications or background processes. • Insufficient RAM for the workload. • Low available storage space. • Malware or unwanted software. • Overheating or thermal throttling. • A failing or heavily fragmented storage drive.
Diagnostic Procedure	1. Check Task Manager for CPU, memory, disk, and network usage. 2. Review startup applications and background processes. 3. Check available storage space. 4. Inspect system temperature and cooling when overheating is suspected. 5. Run a trusted security scan. 6. Check the health and performance of the storage drive.

Solution	Disable unnecessary startup programs, close unnecessary processes, free storage space, remove unwanted software, update the operating system and drivers, improve cooling if needed, and replace an unhealthy storage drive when diagnostics confirm hardware failure.
Lesson Learned	Slow performance should be diagnosed using resource usage rather than simply assuming that the computer needs more RAM. Monitoring helps identify whether CPU, memory, storage, heat, or software is responsible.

Case 3 — Website or Web Application Will Not Load

Problem	A website or locally developed web application does not load correctly, shows an error, or displays a blank page.
Possible Causes	• Web server is not running. • Incorrect project configuration or file path. • Database service is stopped or unavailable. • Port conflict or incorrect port. • Programming error in the application. • Browser cache or extension issue.
Diagnostic Procedure	1. Check the exact error message shown by the browser. 2. Confirm that the required development server or web server is running. 3. Check the terminal or server logs for errors. 4. Verify the application URL, port, and project file structure. 5. Test database connectivity if the application requires a database. 6. Use browser developer tools to inspect console and network errors.
Solution	Start the required server, correct configuration or file paths, resolve database connection problems, change a conflicting port when appropriate, fix the identified programming error, and clear cached data only when the evidence indicates a browser-side issue.
Lesson Learned	Error messages and logs are valuable diagnostic evidence. Reading the actual error before changing code makes troubleshooting faster and reduces the risk of introducing additional problems.

Case 4 — Printer Is Not Printing

Problem	A computer sends a document to a printer, but the printer does not print or remains stuck in the print queue.
Possible Causes	• Printer is offline or powered off. • Incorrect printer selected. • Paper or ink/toner problem. • Print queue is stuck. • Printer driver problem. • Network printer connection problem.
Diagnostic Procedure	1. Check power, cables, paper, and printer status. 2. Confirm that the intended printer is selected as the default or target printer. 3. Check the print queue for stuck jobs. 4. Print a test page from the printer or operating system. 5. Check the printer driver and device status. 6. For network printers, verify that the computer can communicate with the printer.
Solution	Restore the printer connection, correct paper or consumable problems, cancel stuck print jobs, restart the print service when appropriate, update or reinstall the correct driver, and reconnect the network printer if necessary.
Lesson Learned	A printing problem may be caused by hardware, software, or networking. Checking the simplest physical causes first can save time before moving to drivers and operating-system troubleshooting.

Case 5 — Laptop Shows Screen Flickering

Problem	A laptop display repeatedly flickers, changes color, briefly goes black, or returns to normal after a few seconds.
Possible Causes	• Graphics driver problem. • Loose or damaged display cable. • Display panel or hardware fault. • Incorrect refresh-rate or display configuration. • Overheating graphics hardware. • Software conflict or problematic application.
Diagnostic Procedure	1. Observe whether the flickering occurs before Windows loads or only after login. 2. Connect an external monitor to determine whether the problem is limited to the laptop display. 3. Update or roll back the graphics driver if a recent driver change is suspected. 4. Check display resolution and refresh-rate settings. 5. Test the computer in Safe Mode when appropriate. 6. If the issue continues across software environments, inspect the display cable, panel, and graphics hardware through qualified repair procedures.
Solution	Correct the display settings, update or roll back the graphics driver when appropriate, remove software conflicts, and seek hardware inspection if the issue continues. A damaged cable or display panel may require physical repair or replacement.
Lesson Learned	Testing with an external monitor and checking when the problem occurs helps separate software problems from hardware problems. A symptom that continues outside the normal operating environment is stronger evidence of a hardware issue.

Overall Problem-Solving Approach

Across these cases, the recommended approach is to first define the symptom, gather evidence, identify possible causes, test the most likely causes in a controlled order, apply the appropriate solution, and then verify that the problem has been resolved. This approach avoids guessing and helps prevent unnecessary changes that could create additional problems.

Conclusion

IT troubleshooting requires both technical knowledge and a structured problem-solving process. The five cases demonstrate that the same general method can be applied to networking, computer performance, web applications, peripherals, and hardware-related problems. A good IT professional does not immediately assume the cause of a problem; instead, they collect evidence, test possible causes, apply an appropriate fix, and verify the result.